

# Jeonghwan Kim

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## Research Interests

### Multimodal Foundation Models & Representation Learning

Vision-Language Models (VLMs), Vision-Language-Action (VLA) Models, Retrieval-Augmented Generation (RAG), Multimodal Representation Learning, Multi-Hop/Open-Domain Question Answering (QA), Numeracy in LLMs, and Model Interpretability and Robustness.

## Education

### University of Illinois at Urbana-Champaign (UIUC)

Ph.D. Candidate in Computer Science (Advisor: [Heng Ji](#) 🔗)

Champaign, USA

Aug 2023 – Present

### KAIST

M.S. in Computer Science (Advisor: [Sung-Hyon Myaeng](#) 🔗)

Thesis Committee: Sung-Hyon Myaeng, Alice Oh, Junho Lim

GPA: 4.13/4.30

Daejeon, South Korea

Feb 2022

### Handong Global University

B.S. in Computer Science & Electrical Eng. (Advisor: [Heeyoul Henry Choi](#) 🔗)

Magna Cum Laude (Overall GPA: 4.11/4.5 | Major GPA: 4.21/4.5)

Pohang, South Korea

Feb 2020

## Experience

### Meta — Research Scientist Intern

Working on multimodal LLMs.

Redmond, USA

May 2026 – Present

### Meta — Research Scientist Intern (Part-time Student Researcher)

Working on Retrieval-Augmented Generation (RAG) on multimodal LLMs.

Redmond, USA

May 2025 – Oct 2025

### Amazon — Applied Scientist Intern

Developed a generalizable multimodal encoder for geospatial applications.

Bellevue, USA

May 2024 – Aug 2024

### UIUC — Graduate Research Assistant

Research on Multimodal LLMs and reasoning.

Champaign, USA

Aug 2023 – Present

### KAIST — Researcher

Developed graph and RAG-based language modeling approaches for QA.

Daejeon, Korea

Mar 2022 – Jul 2023

### Republic of Korea Marine Corps — Sergeant (Honorably Discharged)

Served as a rifleman and interpreter for the ROK-US Combined Marine Command.

Pohang, Korea

Mar 2015 – Dec 2016

## Publications

‘\*’ indicates equal contribution and co-first author

### [1] Mixture of Layers: Dynamic Layer Routing for Visual Reasoning

**Jeonghwan Kim\***, Sofia Stoica\*, Jiwan Chung, Ansel Blume, Hyeonjeong Ha, Zhenhailong Wang, Xin Luna Dong, Heng Ji

**Preprint (Under Review)**

### [2] Pixel-Grounded Retrieval for Knowledgeable Large Multimodal Models

**Jeonghwan Kim**, Renjie Tao, Sanat Sharma, Jiaqi Wang, Kai Sun, Zhaojiang Lin, Seungwhan Moon, Lambert Mathias, Anuj Kumar, Heng Ji, Xin Luna Dong

[3] **Training VLAs with Vision-Language Anchoring and Language-Action Alignment**

Dwip Dalal, Shivansh Patel, **Jeonghwan Kim**, Utkarsh Mishra, Alex Baratian, Hyeonjeong Ha, Heng Ji, Svetlana Lazebnik, Unnat Jain

Preprint (Under Review)

[4] **Constructive Distortion: Improving MLLMs with Attention-Guided Image Warping**

Dwip Dalal, Gautam Vashishtha, Utkarsh Mishra, **Jeonghwan Kim**, Madhav Kanda, Hyeonjeong Ha, Svetlana Lazebnik, Heng Ji, Unnat Jain

ICLR 2026

[5] **PEARL: Self-Evolving Assistant for Time Management with Reinforcement Learning**

Bingxuan Li, **Jeonghwan Kim**, Cheng Qian, Xiusi Chen, Eitan Anzenberg, Niranjana Kundapur, Heng Ji

ACL 2026

[6] **MM-PoisonRAG: Disrupting Multimodal RAG with Local and Global Knowledge Poisoning Attacks**

Hyeonjeong Ha\*, Qiusi Zhan\*, **Jeonghwan Kim**, Dimitrios Bralios, Saikrishna sanniboina, Nanyun Peng, Kai-Wei Chang, Daniel Kang, Heng Ji

ACL 2026

[7] **PARTONOMY: Large Multimodal Models with Part-Level Visual Understanding**

**Jeonghwan Kim\***, Ansel Blume\*, Hyeonjeong Ha, Elen Chatikyan, Xiaomeng Jin, Khanh Duy Nguyen, Nanyun Peng, Kai-Wei Chang, Derek Hoiem and Heng Ji

NeurIPS 2025 (Spotlight)

[8] **Infogent: An Agent-based Framework for Web Information Aggregation**

**Jeonghwan Kim\***, Revanth Gangi Reddy\*, Sagnik Mukherjee\*, Zhenhailong Wang\*, Dilek Hakkani-Tur, Heng Ji

NAACL 2025, Findings

[9] **SYNTHIA: Novel Concept Design with Affordance Composition**

Hyeonjeong Ha, Xiaomeng Jin, **Jeonghwan Kim**, Jiateng Liu, Zhenhailong Wang, Khanh Duy Nguyen, Ansel Blume, Nanyun Peng, Kai-Wei Chang, Heng Ji

ACL 2025

[10] **Search and Detect: Training-Free Long Tail Object Detection via Web-Image Retrieval**

Mankeerat Sidhu, Hetarth Chopra, Ansel Blume, **Jeonghwan Kim**, Revanth Gangi Reddy, Heng Ji

CVPR 2025

[11] **Aligning LLMs with Individual Preferences via Interaction**

Shujin Wu, May Fung, Cheng Qian, **Jeonghwan Kim**, Dilek Hakkani-Tur, Heng Ji

COLING 2025

[12] **Finer: Investigating and Enhancing Fine-Grained Visual Concept Recognition in Large Vision Language Models**

**Jeonghwan Kim**, Heng Ji

EMNLP 2024

[13] **Why So Gullible? Enhancing the Robustness of Retrieval-Augmented Models against Counterfactual Noise**

**Jeonghwan Kim\***, Giwon Hong\*, Junmo Kang\*, Sung-Hyon Myaeng, Joyce Jiyoung Whang

NAACL 2024, Findings

[14] **ARMADA: Attribute-Based Multimodal Data Augmentation**

Xiaomeng Jin, **Jeonghwan Kim**, Yu Zhou, Kuan-Hao Huang, Te-Lin Wu, Nanyun Peng, Heng Ji

Advancing Natural Language Process for Wikipedia@EMNLP 2024

[15] **MIRACLE: An Online, Explainable Multimodal Interactive Concept Learning System**

Ansel Blume\*, Khanh Duy Nguyen\*, Zhenhailong Wang, Yangyi Chen, Michal Shlapentokh-Rothman, Xiaomeng Jin, **Jeonghwan Kim**, Zhen Zhu, Jiateng Liu, Kuan-Hao Huang, Mankeerat Sidhu, Xuanming Zhang, Vivian Liu, Raunak Sinha, Te-Lin Wu, Abhay Zala, Elias Stengel-Eskin, Da Yin, Yao Xiao, Utkarsh Mall, Zhou Yu, Kai-Wei Chang, Camille Cobb, Karrie Karahalios, Lydia Chilton, Mohit Bansal, Nanyun Peng, Carl Vondrick, Derek Hoiem, Heng Ji

ACM MM Technical Demos, 2024

- [16] **FinePrompt: Unveiling the Role of Finetuned Inductive Bias on Compositional Reasoning in GPT-4**  
*Jeonghwan Kim\**, Giwon Hong\*, Sung-Hyon Myaeng, Joyce Jiyoung Whang  
 EMNLP 2023, Findings
- [17] **Graph-Induced Transformers for Efficient Multi-Hop Question Answering**  
 Giwon Hong, *Jeonghwan Kim*, Junmo Kang, Sung-Hyon Myaeng  
 EMNLP 2022
- [18] **Exploiting Numerical-Contextual Knowledge to Improve Numerical Reasoning in Question Answering**  
*Jeonghwan Kim*, Junmo Kang, Kyung-min Kim, Giwon Hong, Sung-Hyon Myaeng  
 NAACL 2022, Findings
- [19] **Have You Seen That Number? Investigating Extrapolation in Question Answering Models**  
*Jeonghwan Kim*, Giwon Hong, Kyung-min Kim, Junmo Kang, Sung-Hyon Myaeng  
 EMNLP 2021
- [20] **Leveraging Order-Free Tag Relations for Context-Aware Recommendation**  
 Junmo Kang, *Jeonghwan Kim*, Suwon Shin, Sung-Hyon Myaeng  
 EMNLP 2021
- [21] **Can You Distinguish Truthful from Fake Reviews? User Analysis and Assistance Tool for Fake Review Detection**  
*Jeonghwan Kim\**, Junmo Kang\*, Suwon Shin\*, Sung-Hyon Myaeng  
 HCI+NLP Workshop, EACL 2021
- [22] **Maximizing Efficiency of Language Model Pre-training for Learning Representation**  
*Jeonghwan Kim\**, Junmo Kang\*, Suwon Shin\*, Jaeyoung Jo\*, Sung-Hyon Myaeng  
 KSC 2020
- [23] **Object Classification on Raw Radar Data Using Convolutional Neural Networks**  
 Heejae Han, *Jeonghwan Kim*, Junyoung Park, YuJin Lee, Hyunwoo Jo, Yonghyeon Park, Eric T Matson, Seongha Park  
 IEEE SAS 2019
- [24] **Towards the Development and Realization of an Undetectable Stealth UAV**  
 Jiyeon Oh, Daeun Choe, Chanhui Yun, *Jeonghwan Kim*, Michael Hopmeier  
 IEEE IRC 2019

## Academic Services

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**Reviewer** - NeurIPS 2026, ICML 2026, ECCV 2026, CVPR 2026, ACL 2026, EACL 2026, SEA@NeurIPS 2025, AI4Research@AAAI 2025, EMNLP 2025, ACL 2025, ACL-SRW 2025, AAAI 2025, ACL 2024, Language + Molecules@ACL 2024, KnowledgeLM@ACL 2024, EMNLP 2024, EMNLP 2023

## Awards

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- **2026: Capital One Ph.D. Fellowship**: Capital One Center for Generative AI Safety, Knowledge Systems and Cybersecurity (ASKS) @ UIUC
- **2026: Gold Reviewer - ICML, 2026** (International Conference on Machine Learning)
- **2026: Great Reviewer - ACL Rolling Review**, January, 2026 (Association for Computational Linguistics)
- **2025: NeurIPS 2025 Spotlight Publication** - PARTONOMY: Large Multimodal Models with Part-Level Visual Understanding
- **2020: Graduated with Honors in CSE, Magna Cum Laude** (Handong Global University)
- **2020: Finalist (Top 3)**, COMEUP2020 AI CHAMPIONSHIP (K-Startup)
- **2019: Grand Prize**, Capstone Graduation Project Festival (Handong Global University, CSEE)
- **2018: Certificate of Merit**, IITP Capstone 9 Program (Purdue University) - Selected among 30 students across South Korea
- **2018: Finalist (Top 15)**, NAVER AI Hackathon (NAVER AI RUSH)
- **2017: National Science and Technology Scholarship**, Korea Student Aid Foundation (KOSAF)

## Projects

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| <b>DARPA Environment-driven Conceptual Learning (ECOLE)</b><br><i>Funded by DARPA</i>                                                                                                                   | Aug 2023 – Present  |
| <b>DARPA In the Moment (ITM)</b><br><i>Funded by DARPA</i>                                                                                                                                              | Aug 2023 – Present  |
| <b>Center for Advanced Bioenergy and Bioproducts Innovation (CABBI)</b><br><i>Funded by U.S. Department of Energy</i>                                                                                   | Aug 2024 – Present  |
| <b>Development of Expert Decision-making Support AI</b><br>Development of AI technology for explainable expert decision-making.<br><i>Funded by the Korean Government (Ministry of Science and ICT)</i> | Apr 2022 – Jul 2022 |
| <b>Contextual-Numerical Embedding for Reasoning</b><br>Development of context-number embedding for numerical reasoning in QA.<br><i>Funded by the Korean Government (Ministry of Science and ICT)</i>   | Jul 2021 – Dec 2021 |
| <b>ExoBrain</b><br>Development of Knowledge Evolutionary WiseQA Platform Technology.<br><i>Funded by the Korean Government (Ministry of Science and ICT)</i>                                            | Apr 2020 – Jul 2023 |
| <b>Smart Interaction and Context Association</b><br>Machine Learning for Context Association and Smart Interaction Suggestion.<br><i>Funded by the Korean Government (Ministry of Science and ICT)</i>  | Apr 2020 – Mar 2021 |
| <b>AI-based National Online Petition System</b><br>Development of AI-based national Online Petition System for citizen deliberation.<br><i>Funded by KAISTs Exploratory Research Program</i>            | Apr 2021 – Dec 2021 |
| <b>Purdue University - IITP Capstone</b><br>Raw radar data object detection and stealth UAV design.<br><i>Funded by the Korean Government (IITP)</i>                                                    | Aug 2018 – Dec 2018 |

## Teaching

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- **Aug 2020 – Jun 2021:** Teaching Assistant, Senior Data Science, Samsung SDS
- **Jun 2021 – Jul 2021:** Teaching Assistant, Text Mining, SKKU Global Business School
- **Mar 2021 – Dec 2021:** Teaching Assistant, Text Mining, KAIST
- **Mar 2019 – Jun 2019:** Teaching Assistant, Data Structures, Handong Global University

## Technical Skills

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**Programming:** Python, C, C++, Java

**Frameworks:** Pytorch, TensorFlow, Theano, Docker, Huggingface, OpenCV, Flutter, Firebase, DGL

**Tools:** SpaCy, NLTK, Seaborn, Matplotlib